TED (21) –2091 REVISION 2021

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

FUNDAMENTALS OF POLYMER SCIENCE

MODEL QUESTION PAPER – SET-1

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	What are unsaturated hydrocarbon?	[M1.02]	R
2	The valency of Carbon is __________.	[M1.01]	R
3	What is the minimum functionality required for a monomer?	[M2.02]	R
4	What is polymerization?	[M2.03]	U
5	What isthe role of an inhibitor in chain polymerization?	[M3.03]	U
6	Give an example of Ziegler-Natta catalyst.	[M3.03]	R
7	Name any two polymers that are polymerized using bulk polymerization technique.	[M3.04]	R
8	What is the function of antiozonant in polymers?	[M4.04]	U
9	What is meant by thermal degradation of a polymer?	[M4.01]	U

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Convert the given IUPAC name to its chemical structure a. 1,4-butanediol b. 1,3-Dimethylbenzene	[M1.03]	U
2	What are petrochemicals and give any two examples.	[M1.04]	U
3	What are the differences between monomers and polymers? Give an example for each.	[M1.03]	U

4	Based on the structure, explain the classification of polymers with a suitable example for each.	[M2.04]	U
5	Find out the degree of polymerization of a poly(methyl methacrylate) sample having number average molecular weight of 1,50,000 g/mol. (Molecular weight of Methyl methacrylate monomer is 100 g/mol).	[M3.01]	U
6	What is the difference between addition polymerization and condensation polymerization?	[M3.03]	U
7	What is the role of initiators in free radical polymerization and give one example	[M3.02]	U
8	Point out the steps involved chain polymerization.	[M4.03]	U
9	What is chain end degradation of polymer and name one polymer undergoing the degradation	[M4.01]	U
10	What is the purpose of antioxidants in polymer industry? Give one example	[M4.04]	U

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Explain covalent bond. Discuss the bonding in carbon atom. **OR**	[M1.01]	U
IV	Identify the functional groups of the following and give one example each with molecular formula. a. Alcohol b. Alkyl halide c. Amine d. Aldehyde e. Nitrile f. Carboxylic acid g. Ketone	[M1.02]	U
V	Discuss with one example each, the classification of polymers based on the following. (i) Source (ii) Thermal Response (iii) Mode of Formation. **OR**	[M2.04]	U
VI	Explain Weight average molecular weight of polymers and derive	[M2.03]	U

	its equation.		
VII	Describe the following terms with one example each: a. Repeating unit b. Degree of polymerisation c. Oligomer d. Macromolecule e. Micromolecules f. Bi-functional Molecules g. Tri-Functional Molecules **OR**	[M2.02]	U
VIII	Explain the properties of polymers. List everyday uses of any 5 individual polymers.	[M2.01]	U
IX	Give the advantages and disadvantages of bulk and solution polymerization. **OR**	[M3.04]	U
X	Explain the mechanism of free radical polymerization with an example.	[M3.03]	U
XI	Discuss suspension polymerization technique with one example. **OR**	[M3.04]	U
XII	Compare addition and condensation polymerization reaction.	[M3.01]	U
XIII	Discuss the mechanism of oxidative degradation in polymers. **OR**	[M4.02]	U
XIV	Explain the photo degradation of polymers.	[M4.03]	U

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SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

FUNDAMENTALS OF POLYMER SCIENCE

MODEL QUESTION PAPER – SET-2

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

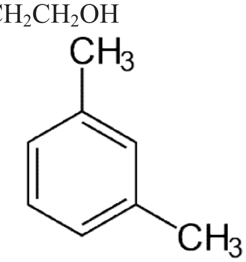
(9 x 1 = 9 Marks)

1	Give any two petrochemicals	[M1.01]	R
2	Draw the structure of glucose molecule.	[M1.03]	R
3	Name any two natural polymers.	[M2.04]	R
4	What is the difference between a monomer and a repeating unit?	[M2.02]	R
5	What are copolymers?	[M3.02]	U
6	What is auto acceleration occurring in bulk polymerization technique?	[M3.04]	U
7	In free radical polymerization, free radicals are produced from the decomposition of ________.	[M3.03]	U
8	What are photostabilizers?	[M4.03]	R
9	Name any two high energy radiations.	[M4.03]	R

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Convert the given chemical structure to its IUPAC name a. $HOCH_2CH_2CH_2CH_2OH$ b. 	[M1.02]	U

2	Explain the type of bonding in NaCl molecule.	[M2.03]	U
3	Explain the following terms and give the equation for both: 1. Degree of Polymerisation 2. Number Average Molecular Weight of Polymer	[M2.04]	U
4	Classify polymers based on their origin with example for each.	[M2.02]	U
5	What are bifunctional monomers? Give 2 Examples.	[M3.04]	U
6	What is solution polymerization technique?	[M3.03]	U
7	What are initiators? How they work?	[M3.01]	U
8	What is addition polymerization? Give two examples of polymer manufactured by this method.	[M4.03]	A
9	What is ultrasonic degradation?	[M4.01]	U
10	Distinguish between chain end and random degradation.	[M4.02]	A

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Compare covalent bond with ionic bond.	[M1.01]	U
	OR		
IV	Compare the following terms. a. Saturated and unsaturated hydrocarbons b. Aliphatic and aromatic hydrocarbons	[M1.02]	U
V	Compare monosaccharide, disaccharide and polysaccharide with an example for each.	[M1.03]	U
	OR		
VI	Explain petroleum and its formation.	[M1.04]	U
VII	Discuss with one example each, the classification of polymers based on the following. (i) Structure (ii) Mode of formation (iii) Thermal response	[M2.04]	
	OR		
VIII	Explain the number average molecular weight of polymers and	[M2.03]	

	derive its equation.		
IX	Explain the types of termination step in free radical polymerization.	[M3.03]	U
	OR		
X	Explain the characters of condensation polymerization reaction.	[M3.01]	U
XI	Discuss suspension polymerization technique.	[M3.04]	U
	OR		
XII	Describe the mechanism of anionic polymerization.	[M3.03]	U
XIII	Explain the significance of anti-oxidants and anti-ozonants.	[M4.04]	U
	OR		
XIV	Explain the biodegradation of polymers.	[M4.02]	A

SCHEME OF VALUATION

FUNDAMENTALS OF POLYMER SCIENCE– SET 1

PART A

I

1. Unsaturated hydrocarbons are hydrocarbons that have double or triple covalent bonds between adjacent carbon atoms.

Eg.Ethene $H_2C = CH_2$

2. Four

3. Two

4. The process of formation of polymers from respective monomers is termed as polymerization.

5. Inhibitors are chemical substance that are capable of inhibiting or killing the chain growth by combining with the active free-radicals and forming either stable products or inactive free-radicals.

6. A mixture of triethyl aluminium and titanium tetrachloride i.e. Al(C2H5)3+TiCl4

7. PMMA and Polystyrene can be polymerized by bulk polymerization technique.

8. An antiozonant is an organic compound that prevents or retards damage caused by ozone.

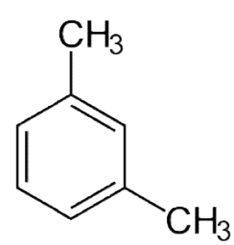
9. The reduction in the physical properties of a polymer caused by changes in its chemical composition as a result of heat is called thermal degradation.

PART B

II

1. a. $HOCH_2CH_2CH_2CH_2OH$

b



2. Petrochemicals are the chemical products obtained from petroleum by refining.

Eg. Ethylene and propylene

Ethylene is used as monomer for the polymerization of Polyethylene and propylene is used as monomer for the polymerization of Polypropylene.

3. Monomers are reactive molecules from which polymer are derived. Polymers are high molecular mass macromolecules, which consists of repeating structural units derived from the monomers.

Examples:

Monomers: Ethylene.

Polymers: Polythene.

4. On the basis of structure polymers are classified as-

(i) Linear polymer: These polymers consist of long and straight chain. Examples : high density polyethylene (HDPE),
(ii) Branched chain polymers: These polymers contain linear chain having some branches. Examples: low density polyethylene (LDPE).
(iii) Cross linked or Network polymers: These polymers have some cross-links between various linear chains.
Example: Bakelite

5. Molecular weigth of polymer, M= 1,50,000 g/mol

Molecular weight of methylmethacrylate monomer, m= 100 g/mol

Degree of polymerization $D_p = \frac{M}{m} = \frac{150000}{100} = 1500$

6. Addition polymerization: It is a polymerization reaction in which monomer units combine without loss of any small molecules such as H2O, NH3, etc. e.g. Preparation of polythene

Condensation polymerization: It is a polymerization reaction in which monomer units combine with loss of small molecules such as H2O, NH3, etc. e.g. preparation of nylon

7. Initiators are thermally unstable compounds and decompose into products called free radicals when energy is supplied into this compounds. These free radicals being very unstable, react with the monomer unit and form a free radical of the monomer. This initate the polymerization reaction.

Cumyl peroxide is an example of initiator

8. The Chain polymerization is characterized by a self-addition of the monomer molecules, to each other, very rapidly through a chain reaction. No by-products are formed and the product has the same elemental composition as that of the monomer. Compounds containing reactive double bonds undergo chain polymerization.

Chain polymerization consist of 3 major steps, initiation, propagation and termination.

9. The chain end degradation starts from the chain ends, resulting in successive release of the monomeric units. This phenomenon is the reverse of propagation step in the chain polymerization. As a result of this degradation, the molecular weight of the polymer decreases slowly and a large quantity of monomer is liberated.

10. Many polymers are protected against oxidative degradation by incorporating certain chemical compounds called antioxidants. Even in small amounts, antioxidants are significantly effective in preserving the original properties of the polymers.

Eg.Phenyl-beta-naphthylimine (PBNA)

PART C

III. A covalent bond is formed by equal sharing of electrons from both the participating atoms. The pair of electrons participating in this type of bonding is called shared pair or bonding pair. The covalent bonds are also termed as molecular bonds. Sharing of bonding pairs will ensure that the atoms achieve stability in their outer shell which is similar to the atoms of noble gases.

As per the electronic configuration of Carbon, it needs to gain or lose 4 electrons to become stable, which seems impossible as:

- Carbon cannot gain 4 electrons to become C4-, because it will be tough for 6 protons to hold 10 electrons and so the atom will become unstable.
- Carbon cannot lose 4 electrons to become C4+ because it would require a large amount of energy to remove out 4 electrons and also the C4+ would have only 2 electrons held by proton, which will again become unstable
- Carbon cannot gain or donate electrons, so to complete its nearest noble gas configuration, it shares electron to form a covalent bond.

IV.

a. Functional Group – R-OH
   Ethanol CH3CH2OH
b. Functional Group – R-X
   Chloroethane CH3CH2Cl
c. Functional Group- RNH_2
   Ethanamine CH3CH2NH2
d. Functional Group- R-CHO
   Ethanal CH3CHO
e. Functional Group- R-CN
   Propanenitrile CH3CH2CN
f. Functional Group- R-COOH
   Ethanoic acid CH3COOH
g. Functional Group- $R_1 - CO - R_2$
   2-Propanone CH3COCH3

V.

i. Based on Source, Polymers can be classified as
   1. Natural Polymers: They are available in nature. Eg- Natural Rubber
   2. Semisynthetic Polymers: They are chemically modified natural polymers. Eg. Halogenated Natural Rubbers
   3. Synthetic Polymers: They are man-made polymers prepared synthetically. Eg-Polyethylene

ii. Based on thermal response, polymers can be classified as
   a. Thermosetting polymers are the class of plastics, which on heating in a mould, becomes infusible and form an insoluble hard mass due to extensive cross-linking between different chains forming three dimensional network bonds. They cannot be used again and again. Examples: Bakelit.
   b. Thermoplastics are linear polymers which are hard at normal temperature and on heating become soft or fluid and hence can be moulded in any shape. They show reversible changes when heated and cooled. Thus, they can be used again and again. Examples: Polythene.

iii. Based on Mode of Formation, polymers can be classified as
   a. Addition polymerization: It is a polymerization reaction in which monomer units combine without loss of any small molecules such as H2O, NH3, etc. e.g. polythene
   b. Condensation polymerization: It is a polymerization reaction in which monomer units combine with loss of small molecules such as H2O, NH3, etc. e.g. nylon

VI. Weight average molecular weight measuring system includes the mass of individual chains, which contributes to the overall molecular weight of the polymer. It is based on the fact that bigger molecules contain more mass than smaller molecules. Thus larger the size of molecule more its contribution to the <u>molecular weight average</u>.

Derivation

Consider a polymer sample with n as the number of molecules and M as the molecular weight .

Let n_1 be a polymer chain with molecular weight M_1, n_2 with M_2 etc

n = $n_1 + n_2 + n_3 + \cdots + n_i = \sum n_i$

M = $M_1 + M_2 + M_3 + \cdots + M_i = \sum M_i$

Weight of chain 1, $W_1 = n_1M_1$

Weight Fraction of chain 1 = $\frac{n_1M_1}{W} = \frac{n_1M_1}{\sum n_iM_i}$

Molecular weigth contribution by chain 1 is given by,

$$\frac{n_1M_1M_1}{\sum n_iM_i} = \frac{n_1{M_1}^2}{\sum n_iM_i}$$

Similarly, the molecular contributions by other chains will be,

$$\frac{n_2{M_2}^2}{\sum n_iM_i}, \frac{n_3{M_3}^2}{\sum n_iM_i}, \frac{n_4{M_4}^2}{\sum n_iM_i}, \cdots \frac{n_i{M_i}^2}{\sum n_iM_i}$$

The weight average molecular weight of the whole polymer will be,

$$\frac{n_2 M_2{}^2}{\sum n_i M_i} + \frac{n_3 M_3{}^2}{\sum n_i M_i} + \frac{n_4 M_4{}^2}{\sum n_i M_i} + \cdots + \frac{n_i M_i{}^2}{\sum n_i M_i} = \frac{\sum n_i M_i{}^2}{\sum n_i M_i} = \overline{M_w}$$

VII.

h. Repeating unit
   A repeat unit or repeating unit is a part of a polymer whose repetition would produce the complete polymer chain by linking the repeat units together successively along the chain
   Eg -($CH_2 - CH_2$-)
i. Degree of Polymerization
   The degree of polymerization is the number of monomeric units in a macromolecule or polymer or oligomer molecule.
j. Oligomer
   Oligomers are low molecular weight polymers comprising a small number of repeat units whose physical properties are significantly dependent on the length of the chain.
   Eg-Liquid paraffin
k. Macromolecule
   Macromolecules are molecule with very high molecular weight.
   Eg – Polyethylene
l. Micromolecules
   Micromolecules are small molecules with a very low molecular weight
   Eg- Water molecule
m. Bi-functional Molecules
   Eg. Molecules with two functional sites are called bifunctional molecules.
   Eg-Ethylene
n. Tri-Functional Molecules
   Molecules with three functional sites are called trifunctional molecules. They result in non linear or branched polymer chains.
   Eg. Glycerin

VIII. **Properties**

Physical Properties

- As chain length and cross-linking increases, the tensile strength of the polymer increases.
- Polymers do not melt, they change state from crystalline to semi-crystalline.

Chemical Properties

- Compared to conventional molecules with different side molecules, the polymer is enabled with hydrogen bonding and ionic bonding resulting in better cross-linking strength.
- Dipole-dipole bonding side chains enable the polymer for high flexibility.

- Polymers with Van der Waals forces linking chains are known to be weak, but give the polymer a low melting point.

Optical Properties

- Due to their ability to change their refractive index with temperature as in the case of PMMA and HEMA: MMA, they are used in lasers for applications in spectroscopy and analytical applications.

Uses

- Polypropene finds usage in a broad range of industries such as textiles, packaging, stationery, plastics, aircraft, construction, rope, toys, etc.
- Polystyrene is one of the most common plastic, actively used in the packaging industry. Bottles, toys, containers, trays, disposable glasses and plates, tv cabinets and lids are some of the daily-used products made up of polystyrene. It is also used as an insulator.
- The most important use of polyvinyl chloride is the manufacture of sewage pipes. It is also used as an insulator in the electric cables.
- Urea-formaldehyde resins are used for making adhesives, moulds, laminated sheets, unbreakable containers, etc.
- Bakelite is used for making electrical switches, kitchen products, toys, jewellery, firearms, insulators, computer discs, etc

IX. Bulk polymerization has several advantages over other methods, these advantages are:

- The system is simple and requires thermal insulation.
- The polymer obtained is pure.
- Large castings may be prepared directly.
- Molecular weight distribution can be easily changed
- The product obtained has high optical clarity

Disadvantages:

- Heat transfer and mixing become difficult as the viscosity of reaction mass increases.
- The problem of heat transfer is compounded by the highly exothermic nature of free radical addition polymerization.
- The polymerization is obtained with a broad molecular weight distribution due to the high viscosity and lack of good heat transfer.
- Very high molecular weights are obtained.
- Gel effect.

Solution polymerization

Advantages:

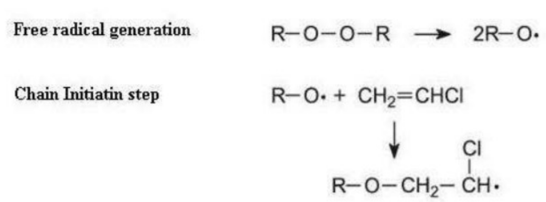
- Heat released by the reaction is absorbed by the solvent, and so the reaction rate is controlled
- The viscosity of the reaction mixture is reduced, not allowing auto-acceleration at high monomer concentrations.
- Decrease of viscosity of reaction mixture by dilution also helps for the heat transfer, one of the major issues connected with polymer production as most of polymerizations are exothermic reactions.

Disadvantages:

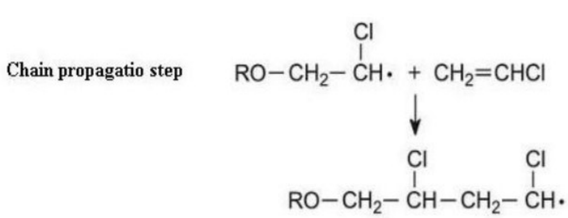
- After the completion of reaction, excess solvent has to be removed in order to obtain the pure polymer.
- decrease of monomer and initiator concentration leading to reduction of reaction rate
- Additional cost of the process related to solvent recycling, toxicity and other environmental impacts of most of organic solvents.

X. Free radical mechanism: It is the way of polymerization where the initiator of the reaction is a free radical. This initiator free radical is produced by many way but one of the important way is decomposition of the compound by sun light, heat, catalyst etc. As example peroxide, hydroperoxide, peracid are very useful initiator in organic polymerization reaction. Decomposition of benzoyl peroxide in presence of sun light or heat form free radical as initiator. Now, the following three common steps of free radical polymerization reaction are:

Chain initiation: This is the first step of polymerization reaction where initiator is added to the monomer unit resulting in the formation of free radical center in monomer unit.

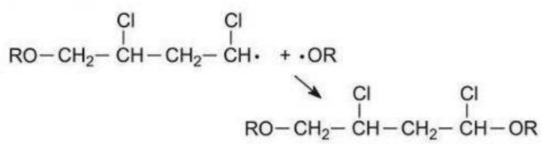


Chain propagation: This is the second rapid step of polymerization reaction where second third forth etc. monomer units are added with generated free radical monomer unit consecutively resulting in the formation of large chain radical.



Chain termination: This is the third and final step of polymerization reaction where termination of free radical after long propagation is takes place by the following possible way

Chain termination step



a. Recombination of two smaller free radicals

b. Recombination of two larger polymeric free radical chain:

c. Recombination of one smaller free radical monomer with a larger free radical

XI.Suspension polymerization is a heterogeneous radical polymerization process that uses mechanical agitation to mix a monomer or mixture of monomers in a liquid phase, such as water, while the monomers polymerize, forming spheres of polymer. The monomer droplets (size of the order 10-1000 μm) are suspended in the liquid phase. The individual monomer droplets can be considered as undergoing bulk polymerization. The liquid phase outside these droplets help in better conduction of heat and thus tempering the increase in temperature.

While choosing a liquid phase for suspension polymerization, we generally prefer low viscosity, high thermal conductivity and low temperature variation of viscosity. The primary advantage of Suspension polymerization over other types of polymerization is that we can get higher degree of polymerization without monomer boil-off. During this process, there is often a possibility of these monomer droplets to come together and stick to each other and thereby creaming occurs in the solution. To prevent this, we often add a protective colloid or do careful stirring of the mixture. One of the most common suspending agents is polyvinyl alcohol (PVA). Usually, the monomer conversion is complete unlike in bulk polymerization, and the initiator used in this is monomer-soluble.

Practically all common themoplastic polymers can be made by this method.

Eg. Polyvinyl chloride

XII.

Addition Polymerization	**Condensation Polymerization**
Monomers must have either a double bond or triple bond	Monomers must have two similar or different functional groups
Produces no by-products	By-products such as ammonia, water and HCl are produced
Addition of monomers results in polymers	Condensation of monomers result in

	polymers
The molecular weight of the resultant polymers is a multiple of monomer’s molecular weight	The molecular weight of the resultant polymer is not a multiple of monomer’s molecular weight
Lewis acids or bases, radical initiators are catalysts in addition polymerization	The catalysts in condensation polymerization are catalysts in condensation polymerization.
Common examples of addition polymerization are PVC, polyethene, Teflon etc.	Common examples of condensation polymerization are nylon, bakelite, silicon, etc.

XIII. Oxidative degradation usually leads to hardening, discolouration as well as surface ages. The case of oxidative degradation of the polymer depends primarily on its structure.Thus, unsaturated polymers such as polyisoprene or polybutadiene containing double bonds are easily attacked by oxygen.

The early stage of oxidative degradation is the formation of free radical sites on the chain backbone by the attack of molecular oxygen, ozone or some free radical produced by the thermal decomposition, of some additives.

$P \rightarrow \dot{P}$

Where P represents polymer chain.

Once the free radical sites are formed, attack by oxygen becomes easier. Oxygen, because of its biradical nature, adds on at the free radical site, forming a peroxide radical.

$\dot{P} + O_2 \rightarrow PO\dot{O}$

The peroxy radical can now attack the neighbouring segment, abstract hydrogen and form a hydroperoxide and a new free radical site.

$PO\dot{O} + P \rightarrow POOH + \dot{P}$

It can also attack a neighbouring double bond.

The hydroperoxide formed can lead to the formation of several new free radical sites.

$POOH \rightarrow P\dot{O} + \dot{O}H$

$\dot{P} + POOH \rightarrow P\dot{O} + POH$

$P\dot{O} + POOH \rightarrow PO\dot{O} + POH$

With so many free radicals formed, their recombination can occur and result in the termination of the chain reactions.

$PO\dot{O} \rightarrow POOP + O_2$

$\dot{P} + \dot{O}H \rightarrow POH$

XIV. Exposure to ultraviolet (UV) radiation may cause the significant degradation of many materials. UV radiation causes photooxidative degradation which results in breaking of the polymer chains, produces free radical and reduces the molecular weight, causing deterioration of mechanical properties and leading to useless materials, after an unpredictable time. Polystyrene (PS), one of the most important material in the modern plastic industry, has been used all over the world, due to its excellent physical properties and low-cost. When polystyrene is subjected to UV irradiation in the presence of air, it undergoes a rapid yellowing and a gradual embrittlement.

Factors causing the photodegradation

Generally, many factors are responsible for causing photodegradation of polymeric materials. They may be divided into two categories:

I)

Internal impurities, which may contain chromophoric groups that are introduced into macromolecules during polymerization processing and storage; they include:

- a-Hydroperoxide.
- b-Carbonyl.
- c-Unsaturated bonds (C=C).
- d-Catalyst residue.
- e-Charge–transfer (CT) complexes with oxygen.

II)

External impurities, which may contain chromophoric groups, are:

- a-Traces of solvents, catalyst, …. etc.
- b-Compounds from a polluted urban atmosphere and smog, e.g. polynuclear hydrocarbons such as naphthalene and anthracene in polypropylene and polybutadiene.
- c-Additives (pigments, dyes, thermal stabilizers, photostabilizers, …. etc.).
- d-Traces of metals and metal oxides from processing equipment and containers, such as Fe, Ni or Cr.

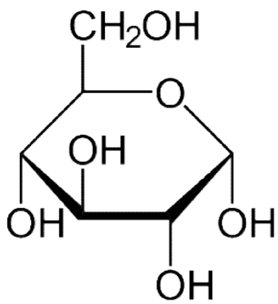
SCHEME OF VALUATION

FUNDAMENTALS OF POLYMER SCIENCE – SET 2

I

1. Ethylene and propylene

2.



3. Starch and cellulose.

4. A monomer is a molecule that forms the basic unit for polymers whereas repeating unit is a part of a polymer whose repetition would produce the complete polymer chain.

5. Copolymer is a polymer derived from more than one species of monomer.

6. Autoacceleration is the process of increasing the rate of reaction in bulk polymerization which may lead to an explosion.

7. Initiators.

8. Photostabilizers protect polymers from degradation occurring due to light.

9. X-rays, Gamma rays

II

1.

c. 1. 1,4-butanediol
d. 1,3-Dimethylbenzene

2. There are primarily three ways in which two atoms combine to lose energy and to become stable. One of the ways is by donating or accepting electrons to complete their octet configuration. The bond formed by this kind of combination is known as an **ionic**

bond or **electrovalent bond**. This kind of bond is formed when one atom gains electrons while the other atom loses electrons from its outermost level or orbit. NaCl have ionic bonding.

3.

i) The degree of polymerization is the number of monomeric units in a polymer or oligomer molecule.

The degree of Polymerisation , Dp = $\frac{Mn}{Mo}$ where M_n is the number-average molecular weight and M_0 is the molecular weight of the monomer unit

ii) The number average molecular weight is defined as the total weight of polymer divided by the total number of molecules.

$\overline{Mn}$= $\frac{\sum n_i M_i}{\sum n_i}$, where $\overline{Mn}$ is the number average molecular weight, n is the number of molecules and M is the molecular weight of each polymer molecule.

4. Polymers can be classified into 3 based on their origin

Natural Polymers:

They occur naturally and are found in plants and animals. For example proteins, starch, cellulose, and rubber. To add up, we also have biodegradable polymers which are called biopolymers.

Semi-synthetic Polymers:

They are derived from naturally occurring polymers and undergo further chemical modification. For example, cellulose nitrate, cellulose acetate.

Synthetic Polymers:

These are man-made polymers. Plastic is the most common and widely used synthetic polymer. It is used in industries and various dairy products. For example, nylon-6, 6, polyether's etc.

5. Molecules with two functional sites are called bifunctional molecules.

Eg-Ethylene and propylene

6. Solution polymerization is a method of industrial polymerization. In this procedure, a monomer is dissolved in a non-reactive solvent that contains a catalyst or initiator.

The reaction results in a polymer which is also soluble in the chosen solvent. Heat released by the reaction is absorbed by the solvent, and so the reaction rate is reduced. Moreover, the viscosity of the reaction mixture is reduced, not allowing autoacceleration at high monomer concentrations. Decrease of viscosity of reaction mixture by dilution also helps for the heat transfer, one of the major issues connected with polymer production as most of polymerizations are exothermic reactions. Once the maximum or desired conversion is reached, excess solvent has to be removed in order to obtain the pure polymer. Hence, solution polymerization is mainly used for applications where the presence of a solvent is desired anyway, as is the case for varnish and adhesives.

7. Initiators are thermally unstable compounds and decompose into products called free radicals when energy is supplied into this compounds. These free radicals being very unstable, react with the monomer unit and form a free radical of the monomer. This initate the polymerization reaction.

Cumyl peroxide is an example of initiator

8. It is a polymerization reaction in which monomer units combine without loss of any small molecules such as H2O, NH3, etc. e.g. polyethylene, polystyrene

9. Reduced in the properties of a polymer due to the exposure to ultrasonic sound is called ultrasonic degradation. It can be considered as a type of mechanical degradation as the polymer sample here will be subjected to very high vibration, which is a mechanical force. The ultrasonic sound passing through the medium induce localized shear gradient and result in chain scission. This reduced the average molecular weight of the polymer.

10. The chain end degradation starts from the chain ends, resulting in successive release of the monomeric units. This phenomenon is the reverse of propagation step in the chain polymerization.

The random degradation occurs at any random point along the polymer chain instead of at the chain ends. This type of degradation can be considered as the reverse of polycondensation process.

III.

IONIC BOND	**COVALENT BOND**
Occurs during the transfer of electrons	Occurs when 2 atoms share their valence electrons
Binding energy is higher than the metallic bond	Binding energy is higher than the metallic bond
Low conductivity	Very low conductivity
Non-directional bond	Directional bond
Present only in one state: solid-state	Present only in all 3 states: solid, liquid, gases
Unmoldable	Unmoldable

Higher melting point	Lower melting point
Non-ductile	Non-ductile
Higher boiling point	Lower boiling point

IV

a.

Saturated Hydrocarbon	**Unsaturated Hydrocarbon**
All carbon atoms are sp^3 hybridized in these compounds.	They contain sp^2 or sp hybridized carbons.
Contain more hydrogen atoms than the corresponding unsaturated hydrocarbons.	Contain fewer hydrogens than the corresponding saturated hydrocarbon.
Examples include alkanes and cycloalkanes.	Examples include alkenes, alkynes, and aromatic hydrocarbons.
They have a relatively low chemical reactivity	They are more reactive than their saturated counterparts.
They generally burn with a blue flame	They generally burn with a sooty flame

b.

Parameter	Aromatic compounds	Aliphatic compounds
Structure	The linking of carbon compounds takes place in the ring structure with the help of conjugated pi electrons	The linking of carbon compounds takes place in a straight line manner
Flame	A sooty flame is	A sooty flame is not produced when burnt

test	produced when burnt	
Odour	Pleasant odour	Unpleasant odour
Example	Benzene and naphthalene	Butane and propane

V

Carbohydrates are biomolecules comprising carbon, hydrogen and oxygen atoms. They are an important source of energy. They are sugars, starch and fibres found in fruits and vegetables.

They simple carbohydrates are of three types.

1. Monosaccharides

Glucose is an example of a carbohydrate monomer or monosaccharide. Other examples of monosaccharides include mannose, galactose, fructose, etc. The structural organization of monosaccharides is as follows:

2. Disaccharides

Two monosaccharides combine to form a disaccharide. Examples of carbohydrates having two monomers include- Sucrose, Lactose, Maltose, etc.

3.Polysaccharides

Polysaccharides are complex carbohydrates formed by the polymerization of a large number of monomers. Examples of polysaccharides include starch, glycogen, cellulose, etc. which exhibit extensive branching and are homopolymers – made up of only glucose units.

VI

Petroleum or mineral oil is the biggest source of energy after coal. It supplies heat and lighting power, machinery lubricants, and raw materials for a variety of manufacturing industries. Petroleum refineries for synthetic textiles, fertilisers and numerous chemical industries act as a nodal industry. Most of India‘s petroleum occurrences are associated with anticlines and fault traps in tertiary-age rock formations. It occurs in folding regions, anticlines, or domes, where oil is trapped in the unfold crest.

Formation of petroleum

- Petroleum is formed from the remains of dead plants and animals.
- When plants and animals die, they sink and settle on the seabed.
- Millions of years ago, these dead wildlife and vegetation decomposed and got mixed with sand and silt.
- Certain bacteria helped in the decomposition of this organic matter and caused some chemical changes.
- Matter consisting of largely carbon and hydrogen was left behind. However, as there is not sufficient oxygen at the bottom of the sea, the matter could not decompose completely.
- The partially decomposed matter remained on the seabed and eventually was covered with multiple layers of sand and silt.
- This burying took millions of years, and finally, due to high temperature and pressure, the organic matter decomposed completely and formed oil.

VII

1. . On the basis of structure polymers are classified as-

(iv) Linear polymer: These polymers consist of long and straight chain. Examples : high density polyethylene (HDPE),

(v) Branched chain polymers: These polymers contain linear chain having some branches. Examples: low density polyethylene (LDPE).

(vi) Cross linked or Network polymers: These polymers have some cross-links between various linear chains.
Example: Bakelite

2. Addition polymerization: It is a polymerization reaction in which monomer units combine without loss of any small molecules such as H2O, NH3, etc. e.g. Preparation of polythene

Condensation polymerization: It is a polymerization reaction in which monomer units combine with loss of small molecules such as H2O, NH3, etc. e.g. preparation of nylon.

3. Based on thermal response, polymers can be classified as

c. Thermosetting polymers are the class of plastics, which on heating in a mould, becomes infusible and form an insoluble hard mass due to extensive cross-linking between different chains forming three dimensional network bonds. They cannot be used again and again. Examples: Bakelit.

d. Thermoplastics are linear polymers which are hard at normal temperature and on heating become soft or fluid and hence can be moulded in any shape. They show reversible changes when heated and cooled. Thus, they can be used again and again. Examples: Polythene.

VIII

The number average molecular weight (M_n) measuring system requires counting the total number of molecules in a unit mass of polymer irrespective of their shape or size. This means all molecules are treated equally.

Derivation

Consider a polymer sample with n as the number of molecules and M as the molecular weight .

Let n_1 be a polymer chain with molecular weight M_1, n_2 with M_2 etc

n = $n_1 + n_2 + n_3 + \cdots + n_i = \sum n_i$

M = $M_1 + M_2 + M_3 + \cdots + M_i = \sum M_i$

Weight of chain 1, $W_1 = n_1 M_1$

Number fraction of chain 1 = $\frac{n_1}{n} = \frac{n_1}{\sum n_i}$

Molecular weigth contribution by chain 1 is given by,

$$= \qquad \frac{n_1 M_1}{\sum n_i}$$

Similarly, the molecular contributions by other chains will be,

$$\frac{n_2 M_2}{\sum n_i}, \frac{n_3 M_3}{\sum n_i}, \frac{n_4 M_4}{\sum n_i}, \cdots \frac{n_i M_i}{\sum n_i}$$

The number average molecular weight of the whole polymer will be,

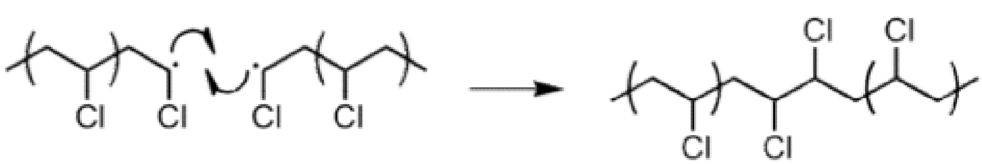
$$\frac{n_1 M_1}{\sum n_i} + \frac{n_2 M_2}{\sum n_i} + \frac{n_3 M_3}{\sum n_i} + \frac{n_4 M_4}{\sum n_i} + \cdots + \frac{n_i M_i}{\sum n_i} = \frac{\sum n_i M_i}{\sum n_i} = \overline{M_n}$$

IX

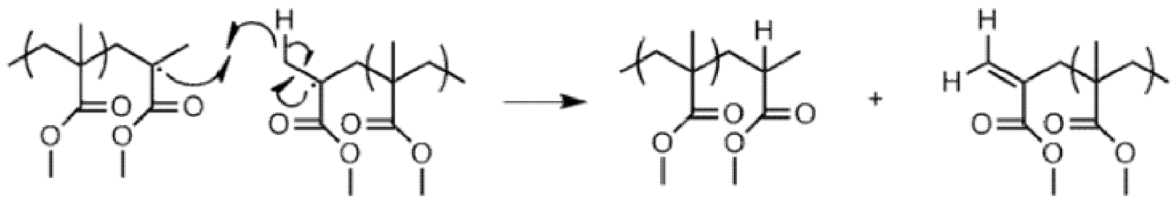
Chain termination is inevitable in radical polymerization due to the high reactivity of radicals. Termination can occur by several different mechanisms. If longer chains are desired, the initiator concentration should be kept low; otherwise, many shorter chains will result.

Combination of two active chain ends: one or both of the following processes may occur.

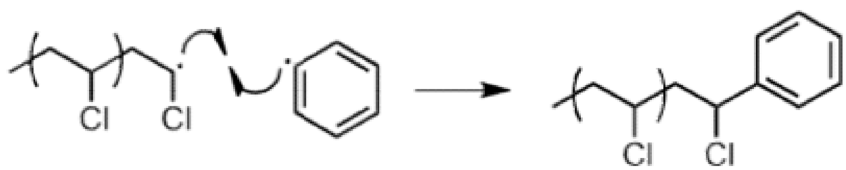
- Combination: two chain ends simply couple together to form one long chain (Figure 14). One can determine if this mode of termination is occurring by monitoring the molecular weight of the propagating species: combination will result in doubling of molecular weight. Also, combination will result in a polymer that is C2 symmetric about the point of the combination.



Radical disproportionation: a hydrogen atom from one chain end is abstracted to another, producing a polymer with a terminal unsaturated group and a polymer with a terminal saturated group



Combination of an active chain end with an initiator radical



X

It is a form of a step-growth polymerization where smaller molecules or monomers react with each other to form larger structural units (usually polymers) while releasing by products such as water or methanol molecule. The by products are normally referred to as condensate.

Characteristics of Condensation Polymerization

Some main characteristics of this type of polymerization are;

- The molecules should have one or two functional groups (like alcohol, amine, or carboxylic acid groups).
- The reaction occurs between two similar or different functional groups or monomers. It can take place between a dimer and oligomer, one monomer and one dimer or between a chain and another chain of polymers.
- Smaller molecules usually combine to form larger molecules.
- Mixed properties of both the molecules or functional groups are taken into consideration.
- A linear polymer is obtained as the condensation product when both functional groups are difunctional.
- When one of the functional groups is tri- or tetra-functional, the polymer formed will be a cross-linked polymer having a three-dimensional network.

- The average molecular weight decreases when monomers are added with one reactive group. Therefore, the functionality of each monomer determines the average molecular weight and cross-link density.

XI

Suspension polymerization is a heterogeneous radical polymerization process that uses mechanical agitation to mix a monomer or mixture of monomers in a liquid phase, such as water, while the monomers polymerize, forming spheres of polymer. The monomer droplets (size of the order 10-1000 μm) are suspended in the liquid phase. The individual monomer droplets can be considered as undergoing bulk polymerization. The liquid phase outside these droplets help in better conduction of heat and thus tempering the increase in temperature.

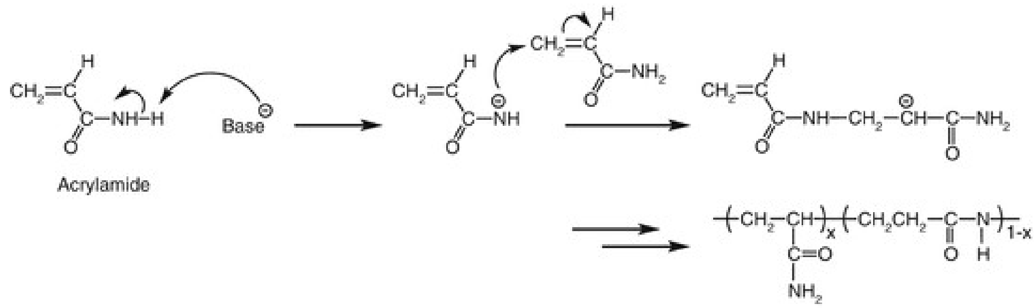
While choosing a liquid phase for suspension polymerization, we generally prefer low viscosity, high thermal conductivity and low temperature variation of viscosity. The primary advantage of Suspension polymerization over other types of polymerization is that we can get higher degree of polymerization without monomer boil-off. During this process, there is often a possibility of these monomer droplets to come together and stick to each other and thereby creaming occurs in the solution. To prevent this, we often add a protective colloid or do careful stirring of the mixture. One of the most common suspending agents is polyvinyl alcohol (PVA). Usually, the monomer conversion is complete unlike in bulk polymerization, and the initiator used in this is monomer-soluble.

This process is used in the production of many commercial resins, including polyvinyl chloride (PVC), a widely used plastic, styrene resins including polystyrene, expanded polystyrene, and high-impact polystyrene, as well as poly(styrene-acrylonitrile) and poly(methyl methacrylate).

XII

It is an addition polymerization that involves the polymerization of monomers that are initiated with anions. This polymerization will be initiated by the transfer of electrons from the ion to the monomer.

The initiators used may be weakly nucleophilic if the monomer is highly electrophilic. In the propagation, the complete consumption of monomer occurs and this will be faster even at low temperature. Generally, vinyl monomers are polymerized by this method. It is very sensitive to the solvent used in the reaction. This method is used for the production of synthetic polydiene rubbers, SBR and thermoplastic styrene elastomers.



XIII

Many polymers are protected against oxidative degradation by incorporating certain chemical compounds called antioxidants. Even in small amounts, antioxidants are significantly effective in preserving the original properties of the polymers.

Eg.Phenyl-beta-naphthylimine (PBNA)

An antiozonant, also known as anti-ozonant, is an organic compound that prevents or retards damage caused by ozone. The most important antiozonants are those which prevent degradation of elastomers like rubber.

Eg.Dialkyl *p*-Phenylenediamines

XIV

Degradation of polymers by the action of living organisms are called biodegradation.

Its advantage is that the polymer will be completely consumed in the environment without the need for complex waste management and that the products of this will be non-toxic. Most common plastics biodegrade very slowly, sometimes to the extent that they are considered non-biodegradable. As polymers are ordinarily too large to be absorbed by microbes biodegradation initially relies on secreted extracellular enzymes to reduce the polymers to manageable chain-lengths. This requires that the polymers bare functional groups that the enzymes are able to recognize, such as ester or amide groups. Long-chain polymers with all-carbon backbones such as polyolefins, polystyrene and PVC will not degrade by biological action alone and must first be oxidised to create chemical groups which the enzymes can attack.

Oxidation can be caused by melt-processing or weathering in the environment, it may also be intentionally accelerated by the addition of biodegradable additives. These are added to the polymer during compounding to improve the biodegradation of otherwise very resistant plastics. Similarly, biodegradable plastics have been designed which are intrinsically biodegradable, provided they are treated like compost and not just left in a landfill site where degradation is very difficult due to the lack of oxygen and moisture.